

8 (a) mass-energy
proton number or charge
nucleon number

B2 [2]

(b) (i) $E_k = \frac{1}{2}mv^2$ and $p = mv$ with working leading to

[via $E_k = \frac{1}{2}m^2v^2/m$ or $\frac{1}{2}m(p/m)^2$]

to $E_k = \frac{p^2}{2m}$

B1 [1]

(ii) $p = (2E_k m)^{1/2}$ hence $(2[E_k m]_\alpha)^{1/2} = (2[E_k m]_{Th})^{1/2}$

C1

$$2 \times [E_k]_{Th} \times 234 = 2 \times 6.69 \times 10^{-13} \times 4$$

C1

$$\begin{aligned} [E_k]_{Th} &= 1.14 \times 10^{-14} \text{ J} \\ &= 71(.5) \text{ keV} \end{aligned}$$

A1

or

calculation of speed of α -particle = $1.42 \times 10^7 \text{ ms}^{-1}$
calculation of momentum of α -particle/nucleus = $9.43 \times 10^{-20} \text{ N s}$

(C1)

$$\begin{aligned} [E_k]_{Th} &= 1.14 \times 10^{-14} \text{ J} \\ &= 71(.5) \text{ keV} \end{aligned}$$

(C1)
(A1) [3]